INFORMATION & COMMUNICATION TECHNOLOGY ACADEMY OF KERALA
ICTAK

INDUSTRIAL CODE QUALITY ASSESSMENT

Software Audit Report

Project: SmartOps

STATUS: NON-COMPLIANT

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Assessment: Automated Static Analysis (SAST)
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User Guide: Understanding Your Audit

1. Executive Summary
Start here for a high-level view of your project's health. Pay close attention to the "Compliance Status".

2. Technical Breakdown
This section lists specific issues found in your code. Each finding includes the file name, line number, and a snippet of the problematic code.

3. Severity Levels
Focus on BLOCKER and CRITICAL issues first. These are high-priority defects that must be resolved before deployment.

4. How to Fix
Use the "Source Context" provided for each finding to locate the issue in your IDE. Follow the "Remediation Roadmap" for a structured approach to improving your score.

Ø<ß“ PRO TIP:

Consistent code quality is a vital skill. Use these reports to learn best practices and avoid recurring mistakes. Re-run the scan after fixing issues to track your progress.

1. Executive Technical Summary

This industrial-grade audit provides a comprehensive evaluation of the source code for "SmartOps". Utilizing industry-standard SonarQube quality profiles, we assessed the codebase for security vulnerabilities, reliability issues, and maintainability concerns.

1.1 Key Performance Indicators (KPIs)

Critical Defects (Bugs)	**1**	*Industry Std: 0*
Security Vulnerabilities	**5**	*Industry Std: 0*
Maintainability (Smells)	**189**	*Target: < 20*
Code Coverage (Unit)	**0%**	*Benchmark: > 80%*
Duplication Density	**3.2%**	*Benchmark: < 3%*
Functional Complexity	**1642**	*Avg/File: < 15*

1.2 Compliance Status

The project fails the industrial quality gate. Critical issues were identified that pose a risk to production stability and security.

2. Technical Finding Breakdown

AUDIT FINDING SQ-0001 CRITICAL

Refactor this function to reduce its Cognitive Complexity from 20 to the 15 allowed.

Location: Backend/src/controllers/aiReviewController.js | Line: 52

CATEGORY: COGNITIVE COMPLEXITY OF FUNCTIONS SHOULD NOT BE TOO HIGH

SOURCE CONTEXT:

```
49 | // ======================================================
  50 | // Accept / Modify AI Suggestion
  51 | // ======================================================
  52 | const reviewAISuggestion = async (req, res) => {
  53 |     try {
  54 |         const { requestId } = req.params;
  55 |
```

AUDIT FINDING SQ-0002 CRITICAL

Refactor this function to reduce its Cognitive Complexity from 26 to the 15 allowed.

Location: Backend/src/controllers/requestController.js | Line: 41

CATEGORY: COGNITIVE COMPLEXITY OF FUNCTIONS SHOULD NOT BE TOO HIGH

SOURCE CONTEXT:

```
38 |     return publicRequest;
  39 | };
  40 |
  41 | const createRequest = async (req, res) => {
  42 |     try {
  43 |         const {
  44 |             title,
```

AUDIT FINDING SQ-0003 CRITICAL

Refactor this function to reduce its Cognitive Complexity from 17 to the 15 allowed.

Location: Backend/src/controllers/requestController.js | Line: 488

CATEGORY: COGNITIVE COMPLEXITY OF FUNCTIONS SHOULD NOT BE TOO HIGH

SOURCE CONTEXT:

```
485 | };
 486 |
 487 |
 488 | const updateRequestStatus = async (req, res) => {
 489 |     try {
 490 |         const { id } = req.params;
 491 |         const { status } = req.body;
```

AUDIT FINDING SQ-0004 CRITICAL

Refactor this function to reduce its Cognitive Complexity from 120 to the 15 allowed.

Location: Frontend/src/layouts/DashboardLayout.jsx | Line: 8

CATEGORY: COGNITIVE COMPLEXITY OF FUNCTIONS SHOULD NOT BE TOO HIGH

SOURCE CONTEXT:

```
5 |
   6 | import "./DashboardLayout.css";
   7 |
   8 | const DashboardLayout = () => {
   9 |     const location = useLocation();
  10 |     const pageMeta = location.pathname.includes("/requests/new")
  11 |         ? { title: "Create Request", subtitle: "Submit a new service request" }
```

AUDIT FINDING SQ-0005 CRITICAL

Refactor this function to reduce its Cognitive Complexity from 23 to the 15 allowed.

Location: Frontend/src/pages/user/RequestDetails.jsx | Line: 26

CATEGORY: COGNITIVE COMPLEXITY OF FUNCTIONS SHOULD NOT BE TOO HIGH

SOURCE CONTEXT:

```
23 |     "Closed"
  24 | ];
  25 |
  26 | const RequestDetails = () => {
  27 |     const navigate = useNavigate();
  28 |     const { id } = useParams();
  29 |     const { user } = useAuth();
```

AUDIT FINDING SQ-0006 **MAJOR**

Simplify this regular expression to reduce its runtime, as it has super-linear performance due to backtracking.

Location: Backend/src/controllers/authController.js | Line: 17

CATEGORY: REGULAR EXPRESSIONS SHOULD NOT CAUSE NON-LINEAR BACKTRACKING

SOURCE CONTEXT:

```
14 |
  15 | const isValidEmail = (email) => (
  16 |     typeof email === "string" &&
  17 |     /^[^\s@]+@[^\s@]+\.[^\s@]+$/.test(email)
  18 | );
  19 |
  20 | const normalizePhone = (phone) => (
```

AUDIT FINDING SQ-0007 **MAJOR**

Extract this nested ternary operation into an independent statement.

Location: Backend/src/controllers/departmentController.js | Line: 225

CATEGORY: TERNARY OPERATORS SHOULD NOT BE NESTED

SOURCE CONTEXT:

```
222 |
 223 |         if (requestCount > 0 || userCount > 0) {
 224 |             return res.status(409).json({
 225 |                 message: `Department cannot be permanently deleted because it is
referenced by ${requestCount} historical request${requestCount === 1 ? "" : "s"}${userCount ?
` and ${userCount} user account${userCount === 1 ? "" : "s"}` : ""}. Deactivate it instead.`
 226 |             });
 227 |         }
 228 |
```

AUDIT FINDING SQ-0008 **MAJOR**

Refactor this code to not use nested template literals.

Location: Backend/src/controllers/departmentController.js | Line: 225

CATEGORY: TEMPLATE LITERALS SHOULD NOT BE NESTED

SOURCE CONTEXT:

```
222 |
 223 |         if (requestCount > 0 || userCount > 0) {
 224 |             return res.status(409).json({
 225 |                 message: `Department cannot be permanently deleted because it is
referenced by ${requestCount} historical request${requestCount === 1 ? "" : "s"}${userCount ?
` and ${userCount} user account${userCount === 1 ? "" : "s"}` : ""}. Deactivate it instead.`
 226 |             });
 227 |         }
 228 |
```

AUDIT FINDING SQ-0009 **MAJOR**

Unnecessary escape character: \".

Location: Backend/src/controllers/requestController.js | Line: 28

CATEGORY: UNNECESSARY CHARACTER ESCAPES SHOULD BE REMOVED

SOURCE CONTEXT:

```
25 |     .replace(/&/g, "&amp;")
  26 |     .replace(/</g, "&lt;")
  27 |     .replace(/>/g, "&gt;")
  28 |     .replace(/\"/g, "&quot;")
  29 |     .replace(/'/g, "&#39;");
  30 |
  31 | const toPublicRequest = (request) => {
```

AUDIT FINDING SQ-0010 **MAJOR**

Simplify this regular expression to reduce its runtime, as it has super-linear performance due to backtracking.

Location: Backend/src/controllers/requestController.js | Line: 177

CATEGORY: REGULAR EXPRESSIONS SHOULD NOT CAUSE NON-LINEAR BACKTRACKING

SOURCE CONTEXT:

```
174 |         try {
 175 |             const recipientEmail = requestCreator?.email?.trim();
 176 |             const hasValidRecipientEmail = Boolean(
 177 |                 recipientEmail && /^[^\s@]+@[^\s@]+\.[^\s@]+$/.test(recipientEmail)
 178 |             );
 179 |
 180 |             console.log("Request creation email event:", {
```

AUDIT FINDING SQ-0011 **MAJOR**

Remove duplicates in this character class.

Location: Backend/src/controllers/requestController.js | Line: 342

CATEGORY: CHARACTER CLASSES IN REGULAR EXPRESSIONS SHOULD NOT CONTAIN THE SAME CHARACTER TWICE

SOURCE CONTEXT:

```
339 |     return (
 340 |         typeof filename === "string" &&
 341 |         path.basename(filename) === filename &&
 342 |         /^[a-zA-Z0-9_-]+\.(jpg|jpeg|png|webp|pdf|doc|docx)$/i.test(
 343 |             filename
 344 |         )
 345 |     );
```

AUDIT FINDING SQ-0012 MAJOR

Prefer using an optional chain expression instead, as it's more concise and easier to read.

Location: Backend/src/controllers/requestController.js | Line: 821

CATEGORY: OPTIONAL CHAINING SHOULD BE PREFERRED

SOURCE CONTEXT:

```
818 |         const { id } = req.params;
 819 |          const { resolution } = req.body;
 820 |
 821 |          if (!resolution || !resolution.trim()) {
 822 |              return res.status(400).json({
 823 |                  message: "Resolution is required"
 824 |              });
```

AUDIT FINDING SQ-0013 MAJOR

Prefer using an optional chain expression instead, as it's more concise and easier to read.

Location: Backend/src/controllers/requestController.js | Line: 1068

CATEGORY: OPTIONAL CHAINING SHOULD BE PREFERRED

SOURCE CONTEXT:

```
1065 |          const { id } = req.params;
1066 |          const { text } = req.body;
1067 |
1068 |          if (!text || !text.trim()) {
1069 |              return res.status(400).json({
1070 |                  message: "Comment text is required"
1071 |              });
```

AUDIT FINDING SQ-0014 MAJOR

Simplify this regular expression to reduce its runtime, as it has super-linear performance due to backtracking.

Location: Backend/src/controllers/userController.js | Line: 19

CATEGORY: REGULAR EXPRESSIONS SHOULD NOT CAUSE NON-LINEAR BACKTRACKING

SOURCE CONTEXT:

```
16 |
  17 | const isValidEmail = (email) => (
  18 |     typeof email === "string" &&
  19 |     /^[^\s@]+@[^\s@]+\.[^\s@]+$/.test(email)
  20 | );
  21 |
  22 | const isValidPhone = (phone) => (
```

AUDIT FINDING SQ-0015 MAJOR

Make sure that using this pseudorandom number generator is safe here.

Location: Backend/src/middleware/uploadMiddleware.js | Line: 27

CATEGORY: PSEUDORANDOM NUMBER GENERATORS (PRNGS) SHOULD NOT BE USED IN SECURITY CONTEXTS

SOURCE CONTEXT:

```
24 |         }
  25 |
  26 |         const uniqueName =
  27 |             `${Date.now()}-${Math.round(Math.random() * 1e9)}` +
  28 |             extension;
  29 |
  30 |         cb(null, uniqueName);
```

AUDIT FINDING SQ-0016 MAJOR

Make sure that using this pseudorandom number generator is safe here.

Location: Backend/src/models/Request.js | Line: 310

CATEGORY: PSEUDORANDOM NUMBER GENERATORS (PRNGS) SHOULD NOT BE USED IN SECURITY CONTEXTS

SOURCE CONTEXT:

```
307 | requestSchema.pre("validate", function () {
 308 |     if (this.isNew && !this.requestNumber) {
 309 |         const timestamp = Date.now();
 310 |         const random = Math.floor(1000 + Math.random() * 9000);
 311 |
 312 |         this.requestNumber = `SO-${timestamp}-${random}`;
 313 |     }
```

AUDIT FINDING SQ-0017 MAJOR

Prefer using an optional chain expression instead, as it's more concise and easier to read.

Location: Backend/src/routes/requestRoutes.js | Line: 41

CATEGORY: OPTIONAL CHAINING SHOULD BE PREFERRED

SOURCE CONTEXT:

```
38 |             });
  39 |         }
  40 |
  41 |         if (req.file && req.file.size === 0) {
  42 |             fs.unlink(req.file.path, () => {});
  43 |
  44 |             return res.status(400).json({
```

AUDIT FINDING SQ-0018 MAJOR

Simplify this regular expression to reduce its runtime, as it has super-linear performance due to backtracking.

Location: Backend/src/services/emailService.js | Line: 15

CATEGORY: REGULAR EXPRESSIONS SHOULD NOT CAUSE NON-LINEAR BACKTRACKING

SOURCE CONTEXT:

```
12 | const sanitizeErrorMessage = (error) => String(
  13 |     error?.message || "Email operation failed"
  14 | )
  15 |     .replace(/[\w.+-]+@[\w.-]+\.[A-Za-z]{2,}/g, "[redacted-email]")
  16 |     .replace(/(pass(word)?\s*[=:]\s*)[^\s]+/gi, "$1[redacted]");
  17 |
  18 | const sendEmail = async ({ to, subject, text, html }) => {
```

AUDIT FINDING SQ-0019 MAJOR

Prefer using an optional chain expression instead, as it's more concise and easier to read.

Location: Backend/src/services/notificationService.js | Line: 96

CATEGORY: OPTIONAL CHAINING SHOULD BE PREFERRED

SOURCE CONTEXT:

```
93 |
  94 |         const notifications = [];
  95 |         for (const admin of admins) {
  96 |             if (actorId && admin._id.toString() === actorId.toString()) {
  97 |                 continue;
  98 |             }
  99 |
```

AUDIT FINDING SQ-0020 MAJOR

Make sure that using this pseudorandom number generator is safe here.

Location: Backend/src/utils/generateRefNo.js | Line: 3

CATEGORY: PSEUDORANDOM NUMBER GENERATORS (PRNGS) SHOULD NOT BE USED IN SECURITY CONTEXTS

SOURCE CONTEXT:

```
1 | const generateRefNo = () => {
   2 |     const year = new Date().getFullYear();
   3 |     const random = Math.floor(100000 + Math.random() * 900000);
   4 |     return `SR-${year}-${random}`;
   5 | };
   6 |
```

AUDIT FINDING SQ-0021 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/components/sidebar/Sidebar.css | Line: 156

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
153 |
 154 | .sidebar-link:hover {
 155 |     background: rgba(255, 255, 255, 0.06);
 156 |     color: #ffffff;
 157 | }
 158 |
 159 | .sidebar-link.active {
```

AUDIT FINDING SQ-0022 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/components/sidebar/Sidebar.css | Line: 161

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
158 |
 159 | .sidebar-link.active {
 160 |     background: rgba(255, 255, 255, 0.12);
 161 |     color: #ffffff;
 162 |     box-shadow: inset 3px 0 0 #dce8ed;
 163 | }
 164 |
```

AUDIT FINDING SQ-0023 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/components/sidebar/Sidebar.css | Line: 274

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
271 |
 272 | .sidebar-logout:hover {
 273 |     background: rgba(255, 255, 255, 0.06);
 274 |     color: #ffffff;
 275 | }
 276 |
 277 | @media (max-width: 900px) {
```

AUDIT FINDING SQ-0024 MAJOR

The object passed as the value prop to the Context provider changes every render. To fix this consider wrapping it in a useMemo hook.

Location: Frontend/src/context/AuthContext.jsx | Line: 77

CATEGORY: REACT CONTEXT PROVIDER VALUES SHOULD HAVE STABLE IDENTITIES

SOURCE CONTEXT:

```
74 |
  75 |     return (
  76 |         <AuthContext.Provider
  77 |             value={{
  78 |                 user,
  79 |                 token,
  80 |                 login,
  81 |                 logout,
  82 |                 updateStoredUser
  83 |             }}
... [truncated for readability]
```

AUDIT FINDING SQ-0025 MAJOR

Extract this nested ternary operation into an independent statement.

Location: Frontend/src/layouts/DashboardLayout.jsx | Line: 12

CATEGORY: TERNARY OPERATORS SHOULD NOT BE NESTED

SOURCE CONTEXT:

```
9 |     const location = useLocation();
  10 |     const pageMeta = location.pathname.includes("/requests/new")
  11 |         ? { title: "Create Request", subtitle: "Submit a new service request" }
  12 |         : location.pathname.includes("/requests/")
  13 |             ? { title: "Request Details", subtitle: "Review request information and
activity" }
  14 |             : location.pathname.includes("/requests")
  15 |                 ? { title: "Requests", subtitle: "Track and manage service requests" }
  16 |                 : location.pathname.includes("/tasks")
  17 |                     ? { title: "Tasks", subtitle: "Organise and complete assigned
work" }
  18 |                     : location.pathname.includes("/notifications")
... [truncated for readability]
```

AUDIT FINDING SQ-0026 MAJOR

Extract this nested ternary operation into an independent statement.

Location: Frontend/src/layouts/DashboardLayout.jsx | Line: 14

CATEGORY: TERNARY OPERATORS SHOULD NOT BE NESTED

SOURCE CONTEXT:

```
11 |         ? { title: "Create Request", subtitle: "Submit a new service request" }
  12 |         : location.pathname.includes("/requests/")
  13 |             ? { title: "Request Details", subtitle: "Review request information and
activity" }
  14 |             : location.pathname.includes("/requests")
  15 |                 ? { title: "Requests", subtitle: "Track and manage service requests" }
  16 |                 : location.pathname.includes("/tasks")
  17 |                     ? { title: "Tasks", subtitle: "Organise and complete assigned
work" }
  18 |                     : location.pathname.includes("/notifications")
  19 |                         ? { title: "Notifications", subtitle: "Stay up to date with
SmartOps activity" }
  20 |                         : location.pathname.includes("/settings")
```

```
... [truncated for readability]
```

AUDIT FINDING SQ-0027 **MAJOR**

Extract this nested ternary operation into an independent statement.

Location: Frontend/src/layouts/DashboardLayout.jsx | Line: 16

CATEGORY: TERNARY OPERATORS SHOULD NOT BE NESTED

SOURCE CONTEXT:

```
13 |             ? { title: "Request Details", subtitle: "Review request information and
activity" }
  14 |             : location.pathname.includes("/requests")
  15 |                 ? { title: "Requests", subtitle: "Track and manage service requests" }
  16 |                 : location.pathname.includes("/tasks")
  17 |                     ? { title: "Tasks", subtitle: "Organise and complete assigned
work" }
  18 |                     : location.pathname.includes("/notifications")
  19 |                         ? { title: "Notifications", subtitle: "Stay up to date with
SmartOps activity" }
  20 |                         : location.pathname.includes("/settings")
  21 |                             ? { title: "Settings", subtitle: "Manage your SmartOps
profile" }
  22 |                             : location.pathname.includes("/staff")
... [truncated for readability]
```

AUDIT FINDING SQ-0028 **MAJOR**

Extract this nested ternary operation into an independent statement.

Location: Frontend/src/layouts/DashboardLayout.jsx | Line: 18

CATEGORY: TERNARY OPERATORS SHOULD NOT BE NESTED

SOURCE CONTEXT:

```
15 |                 ? { title: "Requests", subtitle: "Track and manage service requests" }
  16 |                 : location.pathname.includes("/tasks")
  17 |                     ? { title: "Tasks", subtitle: "Organise and complete assigned
work" }
  18 |                     : location.pathname.includes("/notifications")
  19 |                         ? { title: "Notifications", subtitle: "Stay up to date with
SmartOps activity" }
  20 |                         : location.pathname.includes("/settings")
  21 |                             ? { title: "Settings", subtitle: "Manage your SmartOps
profile" }
  22 |                             : location.pathname.includes("/staff")
  23 |                                 ? { title: "Staff Management", subtitle: "Manage
support staff and account status" }
  24 |                                 : location.pathname.includes("/users")
... [truncated for readability]
```

AUDIT FINDING SQ-0029 MAJOR

Extract this nested ternary operation into an independent statement.

Location: Frontend/src/layouts/DashboardLayout.jsx | Line: 20

CATEGORY: TERNARY OPERATORS SHOULD NOT BE NESTED

SOURCE CONTEXT:

```
17 |                     ? { title: "Tasks", subtitle: "Organise and complete assigned work" }
  18 |                     : location.pathname.includes("/notifications")
  19 |                         ? { title: "Notifications", subtitle: "Stay up to date with
SmartOps activity" }
  20 |                         : location.pathname.includes("/settings")
  21 |                             ? { title: "Settings", subtitle: "Manage your SmartOps
profile" }
  22 |                             : location.pathname.includes("/staff")
  23 |                                 ? { title: "Staff Management", subtitle: "Manage
support staff and account status" }
  24 |                                 : location.pathname.includes("/users")
  25 |                                     ? { title: "User Accounts", subtitle: "Manage
regular employee accounts" }
  26 |                                     : location.pathname.includes("/departments")
... [truncated for readability]
```

AUDIT FINDING SQ-0030 MAJOR

Extract this nested ternary operation into an independent statement.

Location: Frontend/src/layouts/DashboardLayout.jsx | Line: 22

CATEGORY: TERNARY OPERATORS SHOULD NOT BE NESTED

SOURCE CONTEXT:

```
19 |                         ? { title: "Notifications", subtitle: "Stay up to date with
SmartOps activity" }
  20 |                         : location.pathname.includes("/settings")
  21 |                             ? { title: "Settings", subtitle: "Manage your SmartOps
profile" }
  22 |                             : location.pathname.includes("/staff")
  23 |                                 ? { title: "Staff Management", subtitle: "Manage
support staff and account status" }
  24 |                                 : location.pathname.includes("/users")
  25 |                                     ? { title: "User Accounts", subtitle: "Manage
regular employee accounts" }
  26 |                                     : location.pathname.includes("/departments")
  27 |                                         ? { title: "Departments", subtitle: "Manage
request routing departments" }
  28 |                                         : location.pathname.includes("/categories")
... [truncated for readability]
```

AUDIT FINDING SQ-0031 **MAJOR**

Extract this nested ternary operation into an independent statement.

Location: Frontend/src/layouts/DashboardLayout.jsx | Line: 24

CATEGORY: TERNARY OPERATORS SHOULD NOT BE NESTED

SOURCE CONTEXT:

```
21 |                             ? { title: "Settings", subtitle: "Manage your SmartOps
profile" }
  22 |                               : location.pathname.includes("/staff")
  23 |                                   ? { title: "Staff Management", subtitle: "Manage
support staff and account status" }
  24 |                                   : location.pathname.includes("/users")
  25 |                                       ? { title: "User Accounts", subtitle: "Manage
regular employee accounts" }
  26 |                                       : location.pathname.includes("/departments")
  27 |                                           ? { title: "Departments", subtitle: "Manage
request routing departments" }
  28 |                                           : location.pathname.includes("/categories")
  29 |                                               ? { title: "Categories", subtitle: "Manage
request categories" }
  30 |                                               : location.pathname.includes("/reports")
... [truncated for readability]
```

AUDIT FINDING SQ-0032 **MAJOR**

Extract this nested ternary operation into an independent statement.

Location: Frontend/src/layouts/DashboardLayout.jsx | Line: 26

CATEGORY: TERNARY OPERATORS SHOULD NOT BE NESTED

SOURCE CONTEXT:

```
23 |                                 ? { title: "Staff Management", subtitle: "Manage support
staff and account status" }
  24 |                                   : location.pathname.includes("/users")
  25 |                                       ? { title: "User Accounts", subtitle: "Manage
regular employee accounts" }
  26 |                                       : location.pathname.includes("/departments")
  27 |                                           ? { title: "Departments", subtitle: "Manage
request routing departments" }
  28 |                                           : location.pathname.includes("/categories")
  29 |                                               ? { title: "Categories", subtitle: "Manage
request categories" }
  30 |                                               : location.pathname.includes("/reports")
  31 |                                                   ? { title: "Reports", subtitle:
"Review operational performance" }
  32 |                                                   : location.pathname.includes("/
analytics")
... [truncated for readability]
```

AUDIT FINDING SQ-0033 **MAJOR**

Extract this nested ternary operation into an independent statement.

Location: Frontend/src/layouts/DashboardLayout.jsx | Line: 28

CATEGORY: TERNARY OPERATORS SHOULD NOT BE NESTED

SOURCE CONTEXT:

```
25 |                                      ? { title: "User Accounts", subtitle: "Manage
regular employee accounts" }
  26 |                                        : location.pathname.includes("/departments")
  27 |                                            ? { title: "Departments", subtitle: "Manage
request routing departments" }
  28 |                                            : location.pathname.includes("/categories")
  29 |                                                ? { title: "Categories", subtitle: "Manage
request categories" }
  30 |                                                : location.pathname.includes("/reports")
  31 |                                                    ? { title: "Reports", subtitle:
"Review operational performance" }
  32 |                                                    : location.pathname.includes("/
analytics")
  33 |                                                        ? { title: "Analytics", subtitle:
"Explore SmartOps activity" }
  34 |                                                        : location.pathname.includes("/
feedback")
... [truncated for readability]
```

AUDIT FINDING SQ-0034 **MAJOR**

Extract this nested ternary operation into an independent statement.

Location: Frontend/src/layouts/DashboardLayout.jsx | Line: 30

CATEGORY: TERNARY OPERATORS SHOULD NOT BE NESTED

SOURCE CONTEXT:

```
27 |                                          ? { title: "Departments", subtitle: "Manage
request routing departments" }
  28 |                                            : location.pathname.includes("/categories")
  29 |                                                ? { title: "Categories", subtitle: "Manage
request categories" }
  30 |                                                : location.pathname.includes("/reports")
  31 |                                                    ? { title: "Reports", subtitle:
"Review operational performance" }
  32 |                                                    : location.pathname.includes("/
analytics")
  33 |                                                        ? { title: "Analytics", subtitle:
"Explore SmartOps activity" }
  34 |                                                        : location.pathname.includes("/
feedback")
  35 |                                                            ? { title: "Feedback",
subtitle: "Review employee feedback" }
  36 |                                                            : location.pathname.includes("/
audit-logs")
... [truncated for readability]
```

AUDIT FINDING SQ-0035 MAJOR

Extract this nested ternary operation into an independent statement.

Location: Frontend/src/layouts/DashboardLayout.jsx | Line: 32

CATEGORY: TERNARY OPERATORS SHOULD NOT BE NESTED

SOURCE CONTEXT:

```
29 |                                                ? { title: "Categories", subtitle: "Manage
request categories" }
  30 |                                                : location.pathname.includes("/reports")
  31 |                                                    ? { title: "Reports", subtitle:
"Review operational performance" }
  32 |                                                    : location.pathname.includes("/
analytics")
  33 |                                                        ? { title: "Analytics", subtitle:
"Explore SmartOps activity" }
  34 |                                                        : location.pathname.includes("/
feedback")
  35 |                                                            ? { title: "Feedback",
subtitle: "Review employee feedback" }
  36 |                                                            : location.pathname.includes("/
audit-logs")
  37 |                                                                ? { title: "Audit Logs",
subtitle: "Review administrative activity" }
  38 |                                                                :
location.pathname.includes("/ai-review")
... [truncated for readability]
```

AUDIT FINDING SQ-0036 MAJOR

Extract this nested ternary operation into an independent statement.

Location: Frontend/src/layouts/DashboardLayout.jsx | Line: 34

CATEGORY: TERNARY OPERATORS SHOULD NOT BE NESTED

SOURCE CONTEXT:

```
31 |                                                  ? { title: "Reports", subtitle: "Review
operational performance" }
  32 |                                                  : location.pathname.includes("/
analytics")
  33 |                                                      ? { title: "Analytics", subtitle:
"Explore SmartOps activity" }
  34 |                                                      : location.pathname.includes("/
feedback")
  35 |                                                          ? { title: "Feedback",
subtitle: "Review employee feedback" }
  36 |                                                          : location.pathname.includes("/
audit-logs")
  37 |                                                              ? { title: "Audit Logs",
subtitle: "Review administrative activity" }
  38 |                                                              :
location.pathname.includes("/ai-review")
  39 |                                                                  ? { title: "AI
Review", subtitle: "Review AI-assisted request suggestions" }
  40 |                                                                  : { title:
"Dashboard", subtitle: "Smart Operations & Service Management" };
... [truncated for readability]
```

AUDIT FINDING SQ-0037 MAJOR

Extract this nested ternary operation into an independent statement.

Location: Frontend/src/layouts/DashboardLayout.jsx | Line: 36

CATEGORY: TERNARY OPERATORS SHOULD NOT BE NESTED

SOURCE CONTEXT:

```
33 |                                                    ? { title: "Analytics", subtitle:
"Explore SmartOps activity" }
  34 |                                                      : location.pathname.includes("/
feedback")
  35 |                                                          ? { title: "Feedback",
subtitle: "Review employee feedback" }
  36 |                                                          : location.pathname.includes("/
audit-logs")
  37 |                                                              ? { title: "Audit Logs",
subtitle: "Review administrative activity" }
  38 |                                                              :
location.pathname.includes("/ai-review")
  39 |                                                                  ? { title: "AI
Review", subtitle: "Review AI-assisted request suggestions" }
  40 |                                                                  : { title:
"Dashboard", subtitle: "Smart Operations & Service Management" };
  41 |
  42 |     return (
... [truncated for readability]
```

AUDIT FINDING SQ-0038 MAJOR

Extract this nested ternary operation into an independent statement.

Location: Frontend/src/layouts/DashboardLayout.jsx | Line: 38

CATEGORY: TERNARY OPERATORS SHOULD NOT BE NESTED

SOURCE CONTEXT:

```
35 |                                                        ? { title: "Feedback", subtitle:
"Review employee feedback" }
  36 |                                                        : location.pathname.includes("/
audit-logs")
  37 |                                                            ? { title: "Audit Logs",
subtitle: "Review administrative activity" }
  38 |                                                            :
location.pathname.includes("/ai-review")
  39 |                                                                ? { title: "AI
Review", subtitle: "Review AI-assisted request suggestions" }
  40 |                                                                : { title:
"Dashboard", subtitle: "Smart Operations & Service Management" };
  41 |
  42 |     return (
  43 |         <div className="dashboard-layout">
```

AUDIT FINDING SQ-0039 MAJOR

Use <input type="button">, <input type="image">, <input type="reset">, <input type="submit">, or <button> instead of the "button" role to ensure accessibility across all devices.

Location: Frontend/src/pages/Notifications.jsx | Line: 348

CATEGORY: PREFER TAG OVER ARIA ROLE

SOURCE CONTEXT:

```
345 |
 346 |                     {filteredNotifications.map((notification) => (
 347 |
 348 |                         <div
 349 |                             key={notification._id}
 350 |                             className={
 351 |                                 notification.isRead
 352 |                                     ? "notification-card read"
 353 |                                     : "notification-card unread"
 354 |                             }
... [truncated for readability]
```

AUDIT FINDING SQ-0040 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/admin/AdminAIReview.css | Line: 95

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
92 |
  93 | .admin-ai-review-message {
  94 |     background: #ffffff;
  95 |     color: #6e7b84;
  96 |     text-align: center;
  97 | }
  98 |
```

AUDIT FINDING SQ-0041 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/admin/AdminAIReview.css | Line: 145

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
142 | .admin-ai-review-status.pending,
 143 | .admin-ai-decision.pending {
 144 |     background: #f4efe3;
 145 |     color: #8a6b2f;
 146 | }
 147 |
 148 | .admin-ai-review-status.reviewed,
```

AUDIT FINDING SQ-0042 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/admin/AdminAIReview.css | Line: 249

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
246 |     padding: 18px;
 247 |     border: 0;
 248 |     background: #fbfcfc;
 249 |     color: #6e7b84;
 250 | }
 251 |
 252 | .admin-ai-duplicates {
```

AUDIT FINDING SQ-0043 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/admin/AdminAIReview.jsx | Line: 155

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
152 |                     <span className="admin-ai-status-badge">{originalRequest.status}</
span>
 153 |                 </div>
 154 |                 <div className="admin-ai-meta-grid">
 155 |                     <div><label>Request number</
label><strong>{originalRequest.requestNumber || originalRequest._id}</strong></div>
 156 |                     <div><label>Category</label><strong>{originalRequest.category ||
"-"}</strong></div>
 157 |                     <div><label>Department</label><strong>{originalRequest.department
|| "-"}</strong></div>
 158 |                     <div><label>Priority</label><strong>{originalRequest.priority ||
"-"}</strong></div>
```

AUDIT FINDING SQ-0044 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/admin/AdminAIReview.jsx | Line: 156

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
153 |                 </div>
 154 |                 <div className="admin-ai-meta-grid">
 155 |                     <div><label>Request number</
label><strong>{originalRequest.requestNumber || originalRequest._id}</strong></div>
 156 |                     <div><label>Category</label><strong>{originalRequest.category ||
"-"}</strong></div>
 157 |                     <div><label>Department</label><strong>{originalRequest.department
|| "-"}</strong></div>
 158 |                     <div><label>Priority</label><strong>{originalRequest.priority ||
"-"}</strong></div>
 159 |                     <div><label>Created</
label><strong>{formatDate(originalRequest.createdAt)}</strong></div>
```

AUDIT FINDING SQ-0045 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/admin/AdminAIReview.jsx | Line: 157

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
154 |                 <div className="admin-ai-meta-grid">
 155 |                     <div><label>Request number</
label><strong>{originalRequest.requestNumber || originalRequest._id}</strong></div>
 156 |                     <div><label>Category</label><strong>{originalRequest.category ||
"-"}</strong></div>
 157 |                     <div><label>Department</label><strong>{originalRequest.department
|| "-"}</strong></div>
 158 |                     <div><label>Priority</label><strong>{originalRequest.priority ||
"-"}</strong></div>
 159 |                     <div><label>Created</
label><strong>{formatDate(originalRequest.createdAt)}</strong></div>
 160 |                     <div><label>AI analyzed</
label><strong>{formatDate(review.aiAnalyzedAt)}</strong></div>
```

AUDIT FINDING SQ-0046 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/admin/AdminAIReview.jsx | Line: 158

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
155 |                     <div><label>Request number</
label><strong>{originalRequest.requestNumber || originalRequest._id}</strong></div>
 156 |                     <div><label>Category</label><strong>{originalRequest.category ||
"-"}</strong></div>
 157 |                     <div><label>Department</label><strong>{originalRequest.department
|| "-"}</strong></div>
 158 |                     <div><label>Priority</label><strong>{originalRequest.priority ||
"-"}</strong></div>
 159 |                     <div><label>Created</
label><strong>{formatDate(originalRequest.createdAt)}</strong></div>
 160 |                     <div><label>AI analyzed</
label><strong>{formatDate(review.aiAnalyzedAt)}</strong></div>
 161 |                 </div>
```

AUDIT FINDING SQ-0047 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/admin/AdminAIReview.jsx | Line: 159

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
156 |                     <div><label>Category</label><strong>{originalRequest.category ||
"-"}</strong></div>
 157 |                      <div><label>Department</label><strong>{originalRequest.department
|| "-"}</strong></div>
 158 |                      <div><label>Priority</label><strong>{originalRequest.priority ||
"-"}</strong></div>
 159 |                      <div><label>Created</
label><strong>{formatDate(originalRequest.createdAt)}</strong></div>
 160 |                      <div><label>AI analyzed</
label><strong>{formatDate(review.aiAnalyzedAt)}</strong></div>
 161 |                 </div>
 162 |                 <p className="admin-ai-description">{originalRequest.description ||
"No description available."}</p>
```

AUDIT FINDING SQ-0048 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/admin/AdminAIReview.jsx | Line: 160

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
157 |                     <div><label>Department</label><strong>{originalRequest.department
|| "-"}</strong></div>
 158 |                      <div><label>Priority</label><strong>{originalRequest.priority ||
"-"}</strong></div>
 159 |                      <div><label>Created</
label><strong>{formatDate(originalRequest.createdAt)}</strong></div>
 160 |                      <div><label>AI analyzed</
label><strong>{formatDate(review.aiAnalyzedAt)}</strong></div>
 161 |                 </div>
 162 |                 <p className="admin-ai-description">{originalRequest.description ||
"No description available."}</p>
 163 |             </section>
```

AUDIT FINDING SQ-0049 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/admin/AdminAIReview.jsx | Line: 173

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
170 |                     </span>
 171 |                 </div>
 172 |                 <div className="admin-ai-suggestion-grid">
 173 |                     <div><label>AI category</label><strong>{review.aiCategory || "-"}</
strong><small>Confidence score: {review.aiCategoryScore ?? 0}</small></div>
 174 |                     <div><label>AI department</label><strong>{review.aiDepartment ||
"-"}</strong><small>Confidence score: {review.aiDepartmentScore ?? 0}</small></div>
 175 |                     <div><label>AI priority</label><strong>{review.aiPriority || "-"}</
strong><small>Confidence score: {review.aiPriorityScore ?? 0}</small></div>
 176 |                 </div>
```

AUDIT FINDING SQ-0050 **MAJOR**

A form label must be associated with a control.

Location: Frontend/src/pages/admin/AdminAIReview.jsx | Line: 174

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
171 |                 </div>
 172 |                 <div className="admin-ai-suggestion-grid">
 173 |                     <div><label>AI category</label><strong>{review.aiCategory || "-"}</
strong><small>Confidence score: {review.aiCategoryScore ?? 0}</small></div>
 174 |                     <div><label>AI department</label><strong>{review.aiDepartment ||
"-"}</strong><small>Confidence score: {review.aiDepartmentScore ?? 0}</small></div>
 175 |                     <div><label>AI priority</label><strong>{review.aiPriority || "-"}</
strong><small>Confidence score: {review.aiPriorityScore ?? 0}</small></div>
 176 |                 </div>
 177 |                 <div className="admin-ai-edit-grid">
```

AUDIT FINDING SQ-0051 **MAJOR**

A form label must be associated with a control.

Location: Frontend/src/pages/admin/AdminAIReview.jsx | Line: 175

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
172 |                 <div className="admin-ai-suggestion-grid">
 173 |                     <div><label>AI category</label><strong>{review.aiCategory || "-"}</
strong><small>Confidence score: {review.aiCategoryScore ?? 0}</small></div>
 174 |                     <div><label>AI department</label><strong>{review.aiDepartment ||
"-"}</strong><small>Confidence score: {review.aiDepartmentScore ?? 0}</small></div>
 175 |                     <div><label>AI priority</label><strong>{review.aiPriority || "-"}</
strong><small>Confidence score: {review.aiPriorityScore ?? 0}</small></div>
 176 |                 </div>
 177 |                 <div className="admin-ai-edit-grid">
 178 |                     <label>Final category<select value={draft.category}
onChange={(event) => updateDraft("category", event.target.value)}><option value="">Select
category</option>{categories.map((item) => <option key={item._id} value={item.name}
>{item.name}</option>)}{draft.category && !categories.some((item) => item.name ===
draft.category) && <option value={draft.category}>{draft.category}</option>}</select></label>
```

AUDIT FINDING SQ-0052 **MAJOR**

A form label must be associated with a control.

Location: Frontend/src/pages/admin/AdminAIReview.jsx | Line: 207

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
204 |                                 </div>
 205 |                                 {duplicateRequest ? (
 206 |                                     <div className="admin-ai-meta-grid">
 207 |                                         <div><label>Category</
label><strong>{duplicateRequest.category || "-"}</strong></div>
 208 |                                         <div><label>Department</
label><strong>{duplicateRequest.department || "-"}</strong></div>
 209 |                                         <div><label>Priority</
label><strong>{duplicateRequest.priority || "-"}</strong></div>
 210 |                                         <div><label>Status</
label><strong>{duplicateRequest.status || "-"}</strong></div>
```

AUDIT FINDING SQ-0053 **MAJOR**

A form label must be associated with a control.

Location: Frontend/src/pages/admin/AdminAIReview.jsx | Line: 208

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
205 |                                     {duplicateRequest ? (
 206 |                                         <div className="admin-ai-meta-grid">
 207 |                                             <div><label>Category</
label><strong>{duplicateRequest.category || "-"}</strong></div>
 208 |                                             <div><label>Department</
label><strong>{duplicateRequest.department || "-"}</strong></div>
 209 |                                             <div><label>Priority</
label><strong>{duplicateRequest.priority || "-"}</strong></div>
 210 |                                             <div><label>Status</
label><strong>{duplicateRequest.status || "-"}</strong></div>
 211 |                                             <div><label>Created</
label><strong>{formatDate(duplicateRequest.createdAt)}</strong></div>
```

AUDIT FINDING SQ-0054 **MAJOR**

A form label must be associated with a control.

Location: Frontend/src/pages/admin/AdminAIReview.jsx | Line: 209

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
206 |                                         <div className="admin-ai-meta-grid">
 207 |                                             <div><label>Category</
label><strong>{duplicateRequest.category || "-"}</strong></div>
 208 |                                             <div><label>Department</
label><strong>{duplicateRequest.department || "-"}</strong></div>
 209 |                                             <div><label>Priority</
label><strong>{duplicateRequest.priority || "-"}</strong></div>
 210 |                                             <div><label>Status</
label><strong>{duplicateRequest.status || "-"}</strong></div>
 211 |                                             <div><label>Created</
label><strong>{formatDate(duplicateRequest.createdAt)}</strong></div>
 212 |                                         </div>
```

AUDIT FINDING SQ-0055 **MAJOR**

A form label must be associated with a control.

Location: Frontend/src/pages/admin/AdminAIReview.jsx | Line: 210

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
207 |                                             <div><label>Category</
label><strong>{duplicateRequest.category || "-"}</strong></div>
 208 |                                             <div><label>Department</
label><strong>{duplicateRequest.department || "-"}</strong></div>
 209 |                                             <div><label>Priority</
label><strong>{duplicateRequest.priority || "-"}</strong></div>
 210 |                                             <div><label>Status</
label><strong>{duplicateRequest.status || "-"}</strong></div>
 211 |                                             <div><label>Created</
label><strong>{formatDate(duplicateRequest.createdAt)}</strong></div>
 212 |                                         </div>
 213 |                                     ) : <p className="admin-ai-empty">The suspected
request is no longer available.</p>}
```

AUDIT FINDING SQ-0056 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/admin/AdminAIReview.jsx | Line: 211

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
208 |                                           <div><label>Department</
label><strong>{duplicateRequest.department || "-"}</strong></div>
 209 |                                           <div><label>Priority</
label><strong>{duplicateRequest.priority || "-"}</strong></div>
 210 |                                           <div><label>Status</
label><strong>{duplicateRequest.status || "-"}</strong></div>
 211 |                                           <div><label>Created</
label><strong>{formatDate(duplicateRequest.createdAt)}</strong></div>
 212 |                                         </div>
 213 |                                       ) : <p className="admin-ai-empty">The suspected
request is no longer available.</p>}
 214 |                                       <div className="admin-ai-duplicate-footer">
```

AUDIT FINDING SQ-0057 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/admin/AdminAnalytics.css | Line: 159

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
156 | .admin-analytics-table th {
 157 |     padding: 13px 18px;
 158 |     background: #f8f9fa;
 159 |     color: #74808a;
 160 |     font-size: 10px;
 161 |     font-weight: 700;
 162 |     text-align: left;
```

AUDIT FINDING SQ-0058 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/admin/AdminAuditLogs.css | Line: 70

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
67 |     border: 1px solid #e1e6ea;
  68 |     border-radius: 10px;
  69 |     text-align: center;
  70 |     color: #6e7b84;
  71 |     font-size: 13px;
  72 | }
  73 |
```

AUDIT FINDING SQ-0059 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/admin/AdminAuditLogs.css | Line: 88

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
85 |     background: #ffffff;
  86 |     border: 1px solid #e1e6ea;
  87 |     border-radius: 10px;
  88 |     color: #89949c;
  89 |     text-align: center;
  90 | }
  91 |
```

AUDIT FINDING SQ-0060 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/admin/AdminAuditLogs.css | Line: 125

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
122 | .admin-audit-table th {
 123 |     padding: 13px 18px;
 124 |     background: #f8f9fa;
 125 |     color: #74808a;
 126 |     font-size: 10px;
 127 |     font-weight: 700;
 128 |     text-align: left;
```

AUDIT FINDING SQ-0061 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/admin/AdminCategories.css | Line: 134

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
131 |
 132 | .status-inactive {
 133 |     color: #64748B;
 134 |     background: #F1F5F9;
 135 |     padding: 3px 10px;
 136 |     border-radius: 6px;
 137 |     font-size: 12px;
```

AUDIT FINDING SQ-0062 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/admin/AdminCategories.jsx | Line: 165

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
162 |                             {editingId ? "Edit category" : "New category"}
 163 |                         </h3>
 164 |
 165 |                         <label>Name</label>
 166 |                         <input
 167 |                             type="text"
 168 |                             name="name"
```

AUDIT FINDING SQ-0063 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/admin/AdminCategories.jsx | Line: 174

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
171 |                             placeholder="e.g. Network issue"
 172 |                         />
 173 |
 174 |                         <label>Description</label>
 175 |                         <textarea
 176 |                             name="description"
 177 |                             rows="3"
```

AUDIT FINDING SQ-0064 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/admin/AdminDashboard.css | Line: 149

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
146 |
 147 |     background: #f8f9fa;
 148 |
 149 |     color: #74808a;
 150 |
 151 |     font-size: 10px;
 152 |     font-weight: 700;
```

AUDIT FINDING SQ-0065 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/admin/AdminDashboard.css | Line: 249

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
246 | .admin-status-closed {
 247 |     background: #eef0f2;
 248 |
 249 |     color: #66727a;
 250 | }
 251 |
 252 | /* PRIORITY */
```

AUDIT FINDING SQ-0066 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/admin/AdminDashboard.css | Line: 257

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
254 | .admin-priority-low {
 255 |     background: #eef2f3;
 256 |
 257 |     color: #60717b;
 258 | }
 259 |
 260 | .admin-priority-medium {
```

AUDIT FINDING SQ-0067 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/admin/AdminDashboard.css | Line: 290

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
287 |
 288 |     border-radius: 10px;
 289 |
 290 |     color: #6e7b84;
 291 |
 292 |     font-size: 13px;
 293 |
```

AUDIT FINDING SQ-0068 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/admin/AdminDepartments.css | Line: 153

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
150 |     border: 1px solid #e1e6ea;
 151 |     border-radius: 10px;
 152 |     text-align: center;
 153 |     color: #6e7b84;
 154 |     font-size: 13px;
 155 | }
 156 |
```

AUDIT FINDING SQ-0069 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/admin/AdminDepartments.css | Line: 174

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
171 | .admin-departments-table th {
 172 |     padding: 13px 18px;
 173 |     background: #f8f9fa;
 174 |     color: #74808a;
 175 |     font-size: 10px;
 176 |     font-weight: 700;
 177 |     text-align: left;
```

AUDIT FINDING SQ-0070 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/admin/AdminDepartments.jsx | Line: 165

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
162 |                            {editingId ? "Edit department" : "New department"}
 163 |                         </h3>
 164 |
 165 |                         <label>Name</label>
 166 |                         <input
 167 |                             type="text"
 168 |                             name="name"
```

AUDIT FINDING SQ-0071 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/admin/AdminDepartments.jsx | Line: 174

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
171 |                             placeholder="e.g. IT Support"
 172 |                         />
 173 |
 174 |                         <label>Description</label>
 175 |                         <textarea
 176 |                             name="description"
 177 |                             rows="3"
```

AUDIT FINDING SQ-0072 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/admin/AdminFeedback.css | Line: 53

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
50 |     border: 1px solid #e1e6ea;
  51 |     border-radius: 10px;
  52 |     text-align: center;
  53 |     color: #6e7b84;
  54 |     font-size: 13px;
  55 | }
  56 |
```

AUDIT FINDING SQ-0073 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/admin/AdminFeedback.css | Line: 71

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
68 |     background: #ffffff;
  69 |     border: 1px solid #e1e6ea;
  70 |     border-radius: 10px;
  71 |     color: #89949c;
  72 |     text-align: center;
  73 | }
  74 |
```

AUDIT FINDING SQ-0074 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/admin/AdminFeedback.css | Line: 108

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
105 | .admin-feedback-table th {
 106 |     padding: 13px 18px;
 107 |     background: #f8f9fa;
 108 |     color: #74808a;
 109 |     font-size: 10px;
 110 |     font-weight: 700;
 111 |     text-align: left;
```

AUDIT FINDING SQ-0075 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/admin/AdminReports.css | Line: 47

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
44 |     border: 1px solid #e1e6ea;
  45 |     border-radius: 10px;
  46 |     text-align: center;
  47 |     color: #6e7b84;
  48 |     font-size: 13px;
  49 | }
  50 |
```

AUDIT FINDING SQ-0076 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/admin/AdminReports.css | Line: 119

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
116 | .admin-reports-table th {
 117 |     padding: 13px 18px;
 118 |     background: #f8f9fa;
 119 |     color: #74808a;
 120 |     font-size: 10px;
 121 |     font-weight: 700;
 122 |     text-align: left;
```

AUDIT FINDING SQ-0077 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/admin/AdminReports.jsx | Line: 212

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
209 |                         onSubmit={handleDateReport}
 210 |                     >
 211 |                         <div>
 212 |                             <label>From</label>
 213 |                             <input
 214 |                                 type="date"
 215 |                                 value={dateFrom}
```

AUDIT FINDING SQ-0078 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/admin/AdminReports.jsx | Line: 221

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
218 |                         </div>
 219 |
 220 |                         <div>
 221 |                             <label>To</label>
 222 |                             <input
 223 |                                 type="date"
 224 |                                 value={dateTo}
```

AUDIT FINDING SQ-0079 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/admin/AdminRequests.css | Line: 173

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
170 |
 171 |     background: #f8f9fa;
 172 |
 173 |     color: #74808a;
 174 |
 175 |     font-size: 9px;
 176 |     font-weight: 700;
```

AUDIT FINDING SQ-0080 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/admin/AdminRequests.css | Line: 312

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
309 |
 310 | .admin-request-status-closed {
 311 |     background: #eef0f2;
 312 |     color: #66727a;
 313 | }
 314 |
 315 | .admin-request-priority-low {
```

AUDIT FINDING SQ-0081 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/admin/AdminRequests.css | Line: 317

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
314 |
 315 | .admin-request-priority-low {
 316 |     background: #eef2f3;
 317 |     color: #60717b;
 318 | }
 319 |
 320 | .admin-request-priority-medium {
```

AUDIT FINDING SQ-0082 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/admin/AdminRequests.css | Line: 386

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
383 |
 384 |     border-radius: 10px;
 385 |
 386 |     color: #89949c;
 387 |
 388 |     text-align: center;
 389 | }
```

AUDIT FINDING SQ-0083 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/admin/AdminRequests.css | Line: 421

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
418 |
 419 |     border-radius: 10px;
 420 |
 421 |     color: #6e7b84;
 422 |
 423 |     text-align: center;
 424 |
```

AUDIT FINDING SQ-0084 MAJOR

Remove this useless assignment to variable "formatDate".

Location: Frontend/src/pages/admin/AdminRequests.jsx | Line: 263

CATEGORY: UNUSED ASSIGNMENTS SHOULD BE REMOVED

SOURCE CONTEXT:

```
260 |         }
 261 |     };
 262 |
 263 |     const formatDate = (date) => {
 264 |
 265 |         if (!date) {
 266 |             return "-";
```

AUDIT FINDING SQ-0085 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/admin/AdminRequests.jsx | Line: 693

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
690 |
 691 |                             </div>
 692 |
 693 |                             <label>
 694 |                                 Select Staff Member
 695 |                             </label>
 696 |
```

AUDIT FINDING SQ-0086 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/admin/AdminStaff.css | Line: 105

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
102 |
 103 |     background: #f8f9fa;
 104 |
 105 |     color: #74808a;
 106 |
 107 |     font-size: 9px;
 108 |     font-weight: 700;
```

AUDIT FINDING SQ-0087 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/admin/AdminStaff.css | Line: 255

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
252 |
 253 |     text-align: center;
 254 |
 255 |     color: #6e7b84;
 256 |
 257 |     font-size: 13px;
 258 | }
```

AUDIT FINDING SQ-0088 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/admin/AdminStaff.jsx | Line: 494

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
491 |
 492 |                             <div className="admin-staff-modal-body">
 493 |
 494 |                                 <label>
 495 |                                     Name
 496 |                                 </label>
 497 |
```

AUDIT FINDING SQ-0089 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/admin/AdminStaff.jsx | Line: 507

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
504 |                                   required
 505 |                                 />
 506 |
 507 |                                 <label>
 508 |                                     Email
 509 |                                 </label>
 510 |
```

AUDIT FINDING SQ-0090 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/admin/AdminStaff.jsx | Line: 520

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
517 |                                   required
 518 |                                 />
 519 |
 520 |                                 <label>
 521 |                                     Phone Number
 522 |                                 </label>
 523 |
```

AUDIT FINDING SQ-0091 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/admin/AdminStaff.jsx | Line: 532

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
529 |                                   placeholder="Enter phone number"
 530 |                                 />
 531 |
 532 |                                 <label>
 533 |                                     Department
 534 |                                 </label>
 535 |
```

AUDIT FINDING SQ-0092 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/admin/AdminStaff.jsx | Line: 561

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
558 |                                {!editingStaff && (
 559 |
 560 |                                    <>
 561 |                                        <label>
 562 |                                            Password
 563 |                                        </label>
 564 |
```

AUDIT FINDING SQ-0093 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/admin/AdminStaff.jsx | Line: 579

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
576 |                                           minLength={6}
 577 |                                       />
 578 |
 579 |                                       <label>
 580 |                                           Confirm Password
 581 |                                       </label>
 582 |
```

AUDIT FINDING SQ-0094 MAJOR

Extract this nested ternary operation into an independent statement.

Location: Frontend/src/pages/admin/AdminStaff.jsx | Line: 615

CATEGORY: TERNARY OPERATORS SHOULD NOT BE NESTED

SOURCE CONTEXT:

```
612 |                                >
 613 |                                    {saving
 614 |                                        ? "Saving..."
 615 |                                        : editingStaff
 616 |                                            ? "Update Staff"
 617 |                                            : "Create Staff"}
 618 |                                </button>
 619 |
 620 |                            </div>
```

AUDIT FINDING SQ-0095 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/admin/AdminTasks.css | Line: 107

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
104 |
 105 |     background: #f8f9fa;
 106 |
 107 |     color: #74808a;
 108 |
 109 |     font-size: 9px;
 110 |     font-weight: 700;
```

AUDIT FINDING SQ-0096 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/admin/AdminTasks.css | Line: 228

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
225 |
 226 |     border-radius: 10px;
 227 |
 228 |     color: #6e7b84;
 229 |
 230 |     text-align: center;
 231 |
```

AUDIT FINDING SQ-0097 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/admin/AdminTasks.jsx | Line: 427

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
424 |
 425 |                             <div className="admin-task-modal-body">
 426 |
 427 |                                 <label>
 428 |                                     Task Title
 429 |                                 </label>
 430 |
```

AUDIT FINDING SQ-0098 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/admin/AdminTasks.jsx | Line: 440

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
437 |                                    required
 438 |                                />
 439 |
 440 |                                <label>
 441 |                                    Description
 442 |                                </label>
 443 |
```

AUDIT FINDING SQ-0099 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/admin/AdminTasks.jsx | Line: 454

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
451 |                                    required
 452 |                                />
 453 |
 454 |                                <label>
 455 |                                    Related Request
 456 |                                </label>
 457 |
```

AUDIT FINDING SQ-0100 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/admin/AdminTasks.jsx | Line: 490

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
487 |
 488 |                                </select>
 489 |
 490 |                                <label>
 491 |                                    Assign To
 492 |                                </label>
 493 |
```

AUDIT FINDING SQ-0101 **MAJOR**

A form label must be associated with a control.

Location: Frontend/src/pages/admin/AdminTasks.jsx | Line: 540

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
537 |
 538 |                                </select>
 539 |
 540 |                                <label>
 541 |                                    Due Date
 542 |                                </label>
 543 |
```

AUDIT FINDING SQ-0102 **MAJOR**

Extract this nested ternary operation into an independent statement.

Location: Frontend/src/pages/admin/AdminUsers.jsx | Line: 78

CATEGORY: TERNARY OPERATORS SHOULD NOT BE NESTED

SOURCE CONTEXT:

```
75 |                                                  type="button"
  76 |                                                    className={account.isActive ===
false ? "staff-activate-button" : "staff-deactivate-button"}
  77 |                                                    disabled={isCurrent || actionId ===
account._id}
  78 |                                                    title={isCurrent ? "You cannot
deactivate your own account" : account.isActive === false ? "Activate account" : "Deactivate
account"}
  79 |                                                    onClick={() => toggleStatus(account)}
  80 |                                                >
  81 |                                                    {account.isActive === false ?
<FiUserCheck /> : <FiUserX />}
```

AUDIT FINDING SQ-0103 **MAJOR**

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/auth/Login.css | Line: 43

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
40 |     border: 1px solid rgba(255, 255, 255, 0.28);
  41 |     border-radius: 12px;
  42 |     background: rgba(255, 255, 255, 0.14);
  43 |     color: #ffffff;
  44 |     font-size: 24px;
  45 |     font-weight: 700;
  46 | }
```

AUDIT FINDING SQ-0104 MAJOR

Duplicate selector ".login-brand-section", first used at line 25

Location: Frontend/src/pages/auth/Login.css | Line: 303

CATEGORY: SELECTORS SHOULD NOT BE DUPLICATED

SOURCE CONTEXT:

```
300 |
 301 | /* ===== Login left panel: paste at the bottom of Login.css ===== */
 302 |
 303 | .login-brand-section {
 304 |     display: flex;
 305 |     flex-direction: column;
 306 |     justify-content: center;
```

AUDIT FINDING SQ-0105 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/auth/Register.css | Line: 42

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
39 |     border: 1px solid rgba(255, 255, 255, 0.28);
  40 |     border-radius: 12px;
  41 |     background: rgba(255, 255, 255, 0.14);
  42 |     color: #ffffff;
  43 |     font-size: 24px;
  44 |     font-weight: 700;
  45 | }
```

AUDIT FINDING SQ-0106 MAJOR

Duplicate selector ".register-brand-section", first used at line 24

Location: Frontend/src/pages/auth/Register.css | Line: 296

CATEGORY: SELECTORS SHOULD NOT BE DUPLICATED

SOURCE CONTEXT:

```
293 |
 294 | /* ===== Register left panel: paste at the bottom of Register.css ===== */
 295 |
 296 | .register-brand-section {
 297 |     display: flex;
 298 |     flex-direction: column;
 299 |     justify-content: center;
```

AUDIT FINDING SQ-0107 MAJOR

Simplify this regular expression to reduce its runtime, as it has super-linear performance due to backtracking.

Location: Frontend/src/pages/auth/Register.jsx | Line: 79

CATEGORY: REGULAR EXPRESSIONS SHOULD NOT CAUSE NON-LINEAR BACKTRACKING

SOURCE CONTEXT:

```
76 |             return;
  77 |         }
  78 |
  79 |         if (!/^\S+@\S+\.\S+$/.test(email)) {
  80 |             setError("Please enter a valid email address.");
  81 |             return;
  82 |         }
```

AUDIT FINDING SQ-0108 MAJOR

Use <output> instead of the "status" role to ensure accessibility across all devices.

Location: Frontend/src/pages/auth/Register.jsx | Line: 161

CATEGORY: PREFER TAG OVER ARIA ROLE

SOURCE CONTEXT:

```
158 |                     </div>
 159 |
 160 |                     {error && <div className="register-error" role="alert">{error}</
div>}
 161 |                     {success && <div className="register-success"
role="status">{success}</div>}
 162 |
 163 |                     <form className="register-form" onSubmit={handleSubmit}>
 164 |                         <div className="register-form-grid">
```

AUDIT FINDING SQ-0109 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/shared/Settings.css | Line: 135

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
132 |
 133 | .stg-input:disabled {
 134 |     background: #f4f6f7;
 135 |     color: #8a949b;
 136 |     cursor: not-allowed;
 137 | }
 138 |
```

AUDIT FINDING SQ-0110 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/shared/Settings.jsx | Line: 121

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
118 |                         <div className="stg-error">{profileError}</div>
 119 |                     )}
 120 |
 121 |                     <label className="stg-label">Full name</label>
 122 |                     <input
 123 |                         className="stg-input"
 124 |                         type="text"
```

AUDIT FINDING SQ-0111 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/shared/Settings.jsx | Line: 130

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
127 |                         onChange={handleProfileChange}
 128 |                     />
 129 |
 130 |                     <label className="stg-label">Email address</label>
 131 |                     <input
 132 |                         className="stg-input"
 133 |                         type="email"
```

AUDIT FINDING SQ-0112 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/shared/Settings.jsx | Line: 139

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
136 |                         onChange={handleProfileChange}
 137 |                     />
 138 |
 139 |                     <label className="stg-label stg-label--readonly">Role</label>
 140 |                     <input
 141 |                         className="stg-input"
 142 |                         type="text"
```

AUDIT FINDING SQ-0113 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/shared/Settings.jsx | Line: 171

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
168 |                         <div className="stg-error">{passwordError}</div>
 169 |                     )}
 170 |
 171 |                     <label className="stg-label">Current password</label>
 172 |                     <input
 173 |                         className="stg-input"
 174 |                         type="password"
```

AUDIT FINDING SQ-0114 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/shared/Settings.jsx | Line: 180

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
177 |                         onChange={handlePasswordChange}
 178 |                     />
 179 |
 180 |                     <label className="stg-label">New password</label>
 181 |                     <input
 182 |                         className="stg-input"
 183 |                         type="password"
```

AUDIT FINDING SQ-0115 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/shared/Settings.jsx | Line: 189

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
186 |                         onChange={handlePasswordChange}
 187 |                     />
 188 |
 189 |                     <label className="stg-label">Confirm new password</label>
 190 |                     <input
 191 |                         className="stg-input"
 192 |                         type="password"
```

AUDIT FINDING SQ-0116 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/staff/StaffDashboard.css | Line: 153

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
150 |
 151 |     background: #f8f9fa;
 152 |
 153 |     color: #74808a;
 154 |
 155 |     font-size: 10px;
 156 |     font-weight: 700;
```

AUDIT FINDING SQ-0117 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/staff/StaffDashboard.css | Line: 245

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
242 | .staff-status-closed {
 243 |     background: #eef0f2;
 244 |
 245 |     color: #66727a;
 246 | }
 247 |
 248 | .staff-priority-low {
```

AUDIT FINDING SQ-0118 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/staff/StaffDashboard.css | Line: 251

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
248 | .staff-priority-low {
 249 |     background: #eef2f3;
 250 |
 251 |     color: #60717b;
 252 | }
 253 |
 254 | .staff-priority-medium {
```

AUDIT FINDING SQ-0119 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/staff/StaffDashboard.css | Line: 284

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
281 |
 282 |     border-radius: 10px;
 283 |
 284 |     color: #6e7b84;
 285 |
 286 |     font-size: 13px;
 287 |
```

AUDIT FINDING SQ-0120 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/staff/StaffRequests.css | Line: 199

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
196 |
 197 | .staff-request-status-closed {
 198 |     background: #eef0f2;
 199 |     color: #66727a;
 200 | }
 201 |
 202 | .staff-request-priority-low {
```

AUDIT FINDING SQ-0121 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/staff/StaffRequests.css | Line: 204

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
201 |
 202 | .staff-request-priority-low {
 203 |     background: #eef2f3;
 204 |     color: #60717b;
 205 | }
 206 |
 207 | .staff-request-priority-medium {
```

AUDIT FINDING SQ-0122 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/staff/StaffRequests.css | Line: 315

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
312 |
 313 |     border-radius: 10px;
 314 |
 315 |     color: #89949c;
 316 |
 317 |     text-align: center;
 318 | }
```

AUDIT FINDING SQ-0123 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/staff/StaffRequests.css | Line: 350

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
347 |
 348 |     border-radius: 10px;
 349 |
 350 |     color: #6e7b84;
 351 |
 352 |     text-align: center;
 353 |
```

AUDIT FINDING SQ-0124 MAJOR

Duplicate selector ".staff-request-card", first used at line 568

Location: Frontend/src/pages/staff/StaffRequests.css | Line: 584

CATEGORY: SELECTORS SHOULD NOT BE DUPLICATED

SOURCE CONTEXT:

```
581 |         gap: 10px 16px;
 582 |     }
 583 |
 584 |     .staff-request-card {
 585 |         align-items: flex-start;
 586 |
 587 |         flex-direction: column;
```

AUDIT FINDING SQ-0125 MAJOR

Duplicate selector ".staff-request-card-side", first used at line 574

Location: Frontend/src/pages/staff/StaffRequests.css | Line: 590

CATEGORY: SELECTORS SHOULD NOT BE DUPLICATED

SOURCE CONTEXT:

```
587 |         flex-direction: column;
 588 |     }
 589 |
 590 |     .staff-request-card-side {
 591 |         width: 100%;
 592 |
 593 |         align-items: flex-start;
```

AUDIT FINDING SQ-0126 MAJOR

Ambiguous spacing before next element strong

Location: Frontend/src/pages/staff/StaffRequests.jsx | Line: 303

CATEGORY: SPACING BETWEEN INLINE ELEMENTS SHOULD BE EXPLICIT

SOURCE CONTEXT:

```
300 |
 301 |                                     <span>
 302 |                                         Category:
 303 |                                         <strong>
 304 |                                             {request.category}
 305 |                                         </strong>
 306 |                                     </span>
```

AUDIT FINDING SQ-0127 MAJOR

Ambiguous spacing before next element strong

Location: Frontend/src/pages/staff/StaffRequests.jsx | Line: 310

CATEGORY: SPACING BETWEEN INLINE ELEMENTS SHOULD BE EXPLICIT

SOURCE CONTEXT:

```
307 |
 308 |                                     <span>
 309 |                                         Created:
 310 |                                         <strong>
 311 |                                             {formatDate(
 312 |                                                 request.createdAt
 313 |                                             )}
```

AUDIT FINDING SQ-0128 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/staff/StaffRequests.jsx | Line: 462

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
459 |
 460 |                         <div className="staff-modal-body">
 461 |
 462 |                             <label>
 463 |                                 Resolution
 464 |                             </label>
 465 |
```

AUDIT FINDING SQ-0129 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/staff/StaffTasks.css | Line: 231

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
228 |     background: #ffffff;
 229 |     border: 1px solid #e1e6ea;
 230 |     border-radius: 10px;
 231 |     color: #89949c;
 232 |     text-align: center;
 233 | }
 234 |
```

AUDIT FINDING SQ-0130 MAJOR

Extract this nested ternary operation into an independent statement.

Location: Frontend/src/pages/user/CreateRequest.jsx | Line: 320

CATEGORY: TERNARY OPERATORS SHOULD NOT BE NESTED

SOURCE CONTEXT:

```
317 |
 318 |             // Apply matching database category
 319 |             category: matchedCategory
 320 |                 ? typeof matchedCategory === "string"
 321 |                     ? matchedCategory.trim()
 322 |                     : String(matchedCategory.name || "").trim()
 323 |                 : previousData.category,
 324 |
 325 |             // Always apply valid AI priority
```

AUDIT FINDING SQ-0131 MAJOR

Make sure that using this pseudorandom number generator is safe here.

Location: Frontend/src/pages/user/CreateRequest.jsx | Line: 384

CATEGORY: PSEUDORANDOM NUMBER GENERATORS (PRNGS) SHOULD NOT BE USED IN SECURITY CONTEXTS

SOURCE CONTEXT:

```
381 |
 382 |             if (!submissionKeyRef.current) {
 383 |                 submissionKeyRef.current = globalThis.crypto?.randomUUID?.() ||
 384 |                     `request-${Date.now()}-${Math.random().toString(36).slice(2)}`;
 385 |             }
 386 |
 387 |             const requestFields = {
```

AUDIT FINDING SQ-0132 MAJOR

Use <output> instead of the "status" role to ensure accessibility across all devices.

Location: Frontend/src/pages/user/CreateRequest.jsx | Line: 508

CATEGORY: PREFER TAG OVER ARIA ROLE

SOURCE CONTEXT:

```
505 |                     )}
 506 |
 507 |                     {success && (
 508 |                         <div className="form-success" role="status">
 509 |                             {success}
 510 |                         </div>
 511 |                     )}
```

AUDIT FINDING SQ-0133 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/user/MyRequests.css | Line: 89

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
86 |     border: 1px solid #dce2e6;
  87 |     border-radius: 7px;
  88 |
  89 |     color: #89949c;
  90 | }
  91 |
  92 | .request-search svg {
```

AUDIT FINDING SQ-0134 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/user/MyRequests.css | Line: 199

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
196 |
 197 |     background: #f8f9fa;
 198 |
 199 |     color: #74808a;
 200 |
 201 |     font-size: 10px;
 202 |     font-weight: 700;
```

AUDIT FINDING SQ-0135 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/user/MyRequests.css | Line: 270

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
267 |
 268 | .status-pending {
 269 |     background: #f4efe3;
 270 |     color: #8a6b2f;
 271 | }
 272 |
 273 | .status-assigned {
```

AUDIT FINDING SQ-0136 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/user/MyRequests.css | Line: 290

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
287 |
 288 | .status-closed {
 289 |     background: #eef0f2;
 290 |     color: #66727a;
 291 | }
 292 |
 293 | .priority-low {
```

AUDIT FINDING SQ-0137 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/user/MyRequests.css | Line: 295

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
292 |
 293 | .priority-low {
 294 |     background: #eef2f3;
 295 |     color: #60717b;
 296 | }
 297 |
 298 | .priority-medium {
```

AUDIT FINDING SQ-0138 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/user/MyRequests.css | Line: 359

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
356 |
 357 |     border-radius: 10px;
 358 |
 359 |     color: #6e7b84;
 360 |
 361 |     font-size: 13px;
 362 |
```

AUDIT FINDING SQ-0139 MAJOR

Non-interactive elements should not be assigned mouse or keyboard event listeners.

Location: Frontend/src/pages/user/MyRequests.jsx | Line: 364

CATEGORY: NON-INTERACTIVE ELEMENTS SHOULDN'T HAVE EVENT HANDLERS

SOURCE CONTEXT:

```
361 |
 362 |                                                 <div className="request-info">
 363 |
 364 |                                                     <strong
 365 |                                                         className="request-
clickable"
 366 |                                                         onClick={() =>
 367 |                                                             navigate(`/user/
requests/${request._id}`)
 368 |                                                         }
 369 |                                                     >
 370 |                                                         {request.title}
... [truncated for readability]
```

AUDIT FINDING SQ-0140 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/user/RequestDetails.css | Line: 118

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
115 |
 116 | .status-pending {
 117 |     background: #f4efe3;
 118 |     color: #8a6b2f;
 119 | }
 120 |
 121 | .status-assigned {
```

AUDIT FINDING SQ-0141 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/user/RequestDetails.css | Line: 138

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
135 |
 136 | .status-closed {
 137 |     background: #eef0f2;
 138 |     color: #66727a;
 139 | }
 140 |
 141 | .priority-low {
```

AUDIT FINDING SQ-0142 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/user/RequestDetails.css | Line: 143

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
140 |
 141 | .priority-low {
 142 |     background: #eef2f3;
 143 |     color: #60717b;
 144 | }
 145 |
 146 | .priority-medium {
```

AUDIT FINDING SQ-0143 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/user/RequestDetails.css | Line: 298

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
295 |
 296 |     background: #eef1f3;
 297 |
 298 |     color: #98a4ab;
 299 |
 300 |     font-size: 14px;
 301 | }
```

AUDIT FINDING SQ-0144 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/user/RequestDetails.css | Line: 382

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
379 |
 380 |     border-radius: 10px;
 381 |
 382 |     color: #6e7b84;
 383 |
 384 |     font-size: 13px;
 385 |
```

AUDIT FINDING SQ-0145 MAJOR

Duplicate selector ".details-card-header h3", first used at line 186

Location: Frontend/src/pages/user/RequestDetails.css | Line: 482

CATEGORY: SELECTORS SHOULD NOT BE DUPLICATED

SOURCE CONTEXT:

```
479 |     overflow: hidden;
 480 | }
 481 |
 482 | .details-card-header h3 {
 483 |     display: flex;
 484 |     align-items: center;
 485 |     gap: 7px;
```

AUDIT FINDING SQ-0146 MAJOR

Remove duplicates in this character class.

Location: Frontend/src/pages/user/RequestDetails.jsx | Line: 195

CATEGORY: CHARACTER CLASSES IN REGULAR EXPRESSIONS SHOULD NOT CONTAIN THE SAME CHARACTER TWICE

SOURCE CONTEXT:

```
192 |
 193 |     const hasEvidenceFile =
 194 |         typeof request?.resolutionEvidence === "string" &&
 195 |         /^[a-zA-Z0-9_-]+\.(jpg|jpeg|png|webp|pdf|doc|docx)$/i.test(
 196 |             request.resolutionEvidence
 197 |         );
 198 |
```

AUDIT FINDING SQ-0147 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/user/RequestDetails.jsx | Line: 424

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
421 |
 422 |                     <div className="details-card-body">
 423 |                         <div className="detail-item">
 424 |                             <label>Description</label>
 425 |                             <p className="request-description">
 426 |                                 {request.description}
 427 |                             </p>
```

AUDIT FINDING SQ-0148 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/user/RequestDetails.jsx | Line: 432

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
429 |
 430 |                         <div className="detail-row">
 431 |                             <div className="detail-item">
 432 |                                 <label>Category</label>
 433 |                                 <span>{request.category || "-"}</span>
 434 |                             </div>
 435 |
```

AUDIT FINDING SQ-0149 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/user/RequestDetails.jsx | Line: 437

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
434 |                             </div>
 435 |
 436 |                              <div className="detail-item">
 437 |                                  <label>Department</label>
 438 |                                  <span>
 439 |                                      {request.department ||
 440 |                                          request.aiDepartment ||
```

AUDIT FINDING SQ-0150 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/user/RequestDetails.jsx | Line: 448

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
445 |
 446 |                         <div className="detail-row">
 447 |                             <div className="detail-item">
 448 |                                 <label>Location</label>
 449 |                                 <span>{request.location || "-"}</span>
 450 |                             </div>
 451 |
```

AUDIT FINDING SQ-0151 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/user/RequestDetails.jsx | Line: 453

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
450 |                             </div>
 451 |
 452 |                              <div className="detail-item">
 453 |                                  <label>Assigned Staff</label>
 454 |                                  <span>
 455 |                                      {request.assignedTo?.name ||
 456 |                                          "Not assigned"}
```

AUDIT FINDING SQ-0152 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/user/RequestDetails.jsx | Line: 463

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
460 |
 461 |                         <div className="detail-row">
 462 |                             <div className="detail-item">
 463 |                                 <label>Created</label>
 464 |                                 <span>
 465 |                                     {formatDate(request.createdAt, true)}
 466 |                                 </span>
```

AUDIT FINDING SQ-0153 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/user/RequestDetails.jsx | Line: 470

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
467 |                             </div>
 468 |
 469 |                             <div className="detail-item">
 470 |                                 <label>Request Number</label>
 471 |                                 <span>
 472 |                                     {request.requestNumber || request._id}
 473 |                                 </span>
```

AUDIT FINDING SQ-0154 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/user/RequestDetails.jsx | Line: 508

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
505 |
 506 |                         {request.resolution && (
 507 |                             <div className="resolution-box">
 508 |                                 <label>Resolution</label>
 509 |                                 <p>{request.resolution}</p>
 510 |                             </div>
 511 |                         )}
```

AUDIT FINDING SQ-0155 **MAJOR**

A form label must be associated with a control.

Location: Frontend/src/pages/user/RequestDetails.jsx | Line: 515

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
512 |
 513 |                         {request.resolutionEvidence && (
 514 |                             <div className="resolution-box">
 515 |                                 <label>Resolution Evidence</label>
 516 |                                 {hasEvidenceFile ? (
 517 |                                     <button
 518 |                                         type="button"
```

AUDIT FINDING SQ-0156 **MAJOR**

Extract this nested ternary operation into an independent statement.

Location: Frontend/src/pages/user/RequestDetails.jsx | Line: 564

CATEGORY: TERNARY OPERATORS SHOULD NOT BE NESTED

SOURCE CONTEXT:

```
561 |                             const state =
 562 |                                 index < currentIndex
 563 |                                     ? "completed"
 564 |                                     : index === currentIndex
 565 |                                       ? "current"
 566 |                                       : "upcoming";
 567 |
 568 |                             const historyItem = (
 569 |                                 request.statusHistory || []
```

AUDIT FINDING SQ-0157 **MAJOR**

Extract this nested ternary operation into an independent statement.

Location: Frontend/src/pages/user/RequestDetails.jsx | Line: 580

CATEGORY: TERNARY OPERATORS SHOULD NOT BE NESTED

SOURCE CONTEXT:

```
577 |                                 <div className="timeline-icon">
 578 |                                     {state === "completed" ? (
 579 |                                         <FiCheckCircle />
 580 |                                     ) : state === "current" ? (
 581 |                                         <FiClock />
 582 |                                     ) : (
 583 |                                         <span />
 584 |                                     )}
 585 |                                 </div>
 586 |
... [truncated for readability]
```

AUDIT FINDING SQ-0158 MAJOR

Extract this nested ternary operation into an independent statement.

Location: Frontend/src/pages/user/RequestDetails.jsx | Line: 633

CATEGORY: TERNARY OPERATORS SHOULD NOT BE NESTED

SOURCE CONTEXT:

```
630 |                             <p className="empty-detail">
 631 |                                 Loading comments...
 632 |                             </p>
 633 |                         ) : commentsError ? (
 634 |                             <p className="empty-detail comment-error">
 635 |                                 {commentsError}
 636 |                             </p>
 637 |                         ) : comments.length === 0 ? (
 638 |                             <p className="empty-detail">
 639 |                                 No comments yet.
... [truncated for readability]
```

AUDIT FINDING SQ-0159 MAJOR

Extract this nested ternary operation into an independent statement.

Location: Frontend/src/pages/user/RequestDetails.jsx | Line: 637

CATEGORY: TERNARY OPERATORS SHOULD NOT BE NESTED

SOURCE CONTEXT:

```
634 |                             <p className="empty-detail comment-error">
 635 |                                 {commentsError}
 636 |                             </p>
 637 |                         ) : comments.length === 0 ? (
 638 |                             <p className="empty-detail">
 639 |                                 No comments yet.
 640 |                             </p>
 641 |                         ) : (
 642 |                             comments.map((item, index) => (
 643 |                                 <div
... [truncated for readability]
```

AUDIT FINDING SQ-0160 MAJOR

A form label must be associated with a control.

Location: Frontend/src/pages/user/RequestDetails.jsx | Line: 721

CATEGORY: LABEL ELEMENTS SHOULD HAVE A TEXT LABEL AND AN ASSOCIATED CONTROL

SOURCE CONTEXT:

```
718 |                             className="feedback-form"
 719 |                             onSubmit={submitFeedback}
 720 |                         >
 721 |                             <label>Rating</label>
 722 |
 723 |                             <select
 724 |                                 value={feedbackRating}
```

AUDIT FINDING SQ-0161 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/user/UserDashboard.css | Line: 152

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
149 |
 150 |     background: #f8f9fa;
 151 |
 152 |     color: #74808a;
 153 |
 154 |     font-size: 10px;
 155 |     font-weight: 700;
```

AUDIT FINDING SQ-0162 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/user/UserDashboard.css | Line: 226

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
223 |
 224 | .status-pending {
 225 |     background: #f4efe3;
 226 |     color: #8a6b2f;
 227 | }
 228 |
 229 | .status-assigned {
```

AUDIT FINDING SQ-0163 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/user/UserDashboard.css | Line: 246

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
243 |
 244 | .status-closed {
 245 |     background: #eef0f2;
 246 |     color: #66727a;
 247 | }
 248 |
 249 | /* PRIORITY */
```

AUDIT FINDING SQ-0164 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/user/UserDashboard.css | Line: 253

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
250 |
 251 | .priority-low {
 252 |     background: #eef2f3;
 253 |     color: #60717b;
 254 | }
 255 |
 256 | .priority-medium {
```

AUDIT FINDING SQ-0165 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/pages/user/UserDashboard.css | Line: 283

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
280 |
 281 |     border-radius: 10px;
 282 |
 283 |     color: #6e7b84;
 284 |
 285 |     font-size: 13px;
 286 |
```

AUDIT FINDING SQ-0166 MAJOR

Text does not meet the minimal contrast requirement with its background.

Location: Frontend/src/theme.css | Line: 49

CATEGORY: TEXT AND BACKGROUND COLORS SHOULD HAVE SUFFICIENT CONTRAST

SOURCE CONTEXT:

```
46 |
  47 | .sidebar-link:hover {
  48 |     background: rgba(96, 165, 250, 0.16);
  49 |     color: #ffffff;
  50 | }
  51 |
  52 | .dashboard-content {
```

AUDIT FINDING SQ-0167 **MINOR**

`String.raw` should be used to avoid escaping `\`.

Location: Backend/src/ai/aiService.js | Line: 9

CATEGORY: STRING LITERALS WITH ESCAPED BACKSLASHES SHOULD USE `STRING.RAW` TEMPLATE LITERALS

SOURCE CONTEXT:

```
6 |
   7 |
   8 | const escapeRegex = (value) =>
   9 |     value.replace(/[.*+?^${}()|[\]\\]/g, "\\$&");
  10 |
  11 | // Whole-word match; allows simple plurals (printer -> printers)
  12 | const containsKeyword = (text, keyword) => {
```

AUDIT FINDING SQ-0168 **MINOR**

Handle this exception or don't catch it at all.

Location: Backend/src/controllers/authController.js | Line: 96

CATEGORY: EXCEPTIONS SHOULD NOT BE IGNORED

SOURCE CONTEXT:

```
93 |             }
  94 |         });
  95 |
  96 |     } catch (error) {
  97 |         res.status(500).json({
  98 |             message: "Registration failed"
  99 |         });
 100 |     }
 101 | };
 102 |
... [truncated for readability]
```

AUDIT FINDING SQ-0169 **MINOR**

Handle this exception or don't catch it at all.

Location: Backend/src/controllers/authController.js | Line: 161

CATEGORY: EXCEPTIONS SHOULD NOT BE IGNORED

SOURCE CONTEXT:

```
158 |         }
 159 |     });
 160 |
 161 |     } catch (error) {
 162 |     res.status(500).json({
 163 |         message: "Login failed"
 164 |     });
 165 | }
 166 |
 167 | };
... [truncated for readability]
```

AUDIT FINDING SQ-0170 **MINOR**

Handle this exception or don't catch it at all.

Location: Backend/src/controllers/departmentController.js | Line: 78

CATEGORY: EXCEPTIONS SHOULD NOT BE IGNORED

SOURCE CONTEXT:

```
75 |             message: "Active departments fetched successfully",
  76 |             departments
  77 |         });
  78 |     } catch (error) {
  79 |         return res.status(500).json({
  80 |             message: "Failed to fetch active departments"
  81 |         });
  82 |     }
  83 | };
  84 |
... [truncated for readability]
```

AUDIT FINDING SQ-0171 **MINOR**

Prefer `Number.isNaN` over `isNaN`.

Location: Backend/src/controllers/reportController.js | Line: 187

CATEGORY: NUMBER STATIC METHODS AND PROPERTIES SHOULD BE PREFERRED OVER GLOBAL EQUIVALENTS

SOURCE CONTEXT:

```
184 |         const fromDate = new Date(from);
 185 |         const toDate = new Date(to);
 186 |
 187 |         if (isNaN(fromDate.getTime()) || isNaN(toDate.getTime())) {
 188 |             return res.status(400).json({
 189 |                 message: "Invalid date format. Use YYYY-MM-DD"
 190 |             });
```

AUDIT FINDING SQ-0172 **MINOR**

Prefer `Number.isNaN` over `isNaN`.

Location: Backend/src/controllers/reportController.js | Line: 187

CATEGORY: NUMBER STATIC METHODS AND PROPERTIES SHOULD BE PREFERRED OVER GLOBAL EQUIVALENTS

SOURCE CONTEXT:

```
184 |         const fromDate = new Date(from);
 185 |         const toDate = new Date(to);
 186 |
 187 |         if (isNaN(fromDate.getTime()) || isNaN(toDate.getTime())) {
 188 |             return res.status(400).json({
 189 |                 message: "Invalid date format. Use YYYY-MM-DD"
 190 |             });
```

AUDIT FINDING SQ-0173 **MINOR**

Prefer `node:fs` over `fs`.

Location: Backend/src/controllers/requestController.js | Line: 4

CATEGORY: NODE.JS BUILT-IN MODULES SHOULD BE IMPORTED USING THE "NODE:" PROTOCOL

SOURCE CONTEXT:

```
1 | const Request = require("../models/Request");
   2 | const User = require("../models/User");
   3 | const mongoose = require("mongoose");
   4 | const fs = require("fs");
   5 | const path = require("path");
   6 | const createAuditLog = require("../utils/auditLogger");
   7 | const {
```

AUDIT FINDING SQ-0174 **MINOR**

Prefer `node:path` over `path`.

Location: Backend/src/controllers/requestController.js | Line: 5

CATEGORY: NODE.JS BUILT-IN MODULES SHOULD BE IMPORTED USING THE "NODE:" PROTOCOL

SOURCE CONTEXT:

```
2 | const User = require("../models/User");
   3 | const mongoose = require("mongoose");
   4 | const fs = require("fs");
   5 | const path = require("path");
   6 | const createAuditLog = require("../utils/auditLogger");
   7 | const {
   8 |     analyzeRequest,
```

AUDIT FINDING SQ-0175 **MINOR**

Prefer `String#replaceAll()` over `String#replace()`.

Location: Backend/src/controllers/requestController.js | Line: 25

CATEGORY: STRINGS SHOULD USE "REPLACEALL()" INSTEAD OF "REPLACE()" WITH GLOBAL REGEX

SOURCE CONTEXT:

```
22 |     "Your request was created successfully, but the confirmation email could not be
sent.";
  23 |
  24 | const escapeHtml = (value) => String(value ?? "")
  25 |     .replace(/&/g, "&amp;")
  26 |     .replace(/</g, "&lt;")
  27 |     .replace(/>/g, "&gt;")
   28 |     .replace(/\"/g, "&quot;")
```

AUDIT FINDING SQ-0176 **MINOR**

Prefer `String#replaceAll()` over `String#replace()`.

Location: Backend/src/controllers/requestController.js | Line: 26

CATEGORY: STRINGS SHOULD USE "REPLACEALL()" INSTEAD OF "REPLACE()" WITH GLOBAL REGEX

SOURCE CONTEXT:

```
23 |
  24 | const escapeHtml = (value) => String(value ?? "")
  25 |     .replace(/&/g, "&amp;")
  26 |     .replace(/</g, "&lt;")
  27 |     .replace(/>/g, "&gt;")
  28 |     .replace(/\"/g, "&quot;")
  29 |     .replace(/'/g, "&#39;");
```

AUDIT FINDING SQ-0177 **MINOR**

Prefer `String#replaceAll()` over `String#replace()`.

Location: Backend/src/controllers/requestController.js | Line: 27

CATEGORY: STRINGS SHOULD USE "REPLACEALL()" INSTEAD OF "REPLACE()" WITH GLOBAL REGEX

SOURCE CONTEXT:

```
24 | const escapeHtml = (value) => String(value ?? "")
  25 |     .replace(/&/g, "&amp;")
  26 |     .replace(/</g, "&lt;")
  27 |     .replace(/>/g, "&gt;")
  28 |     .replace(/\"/g, "&quot;")
  29 |     .replace(/'/g, "&#39;");
  30 |
```

AUDIT FINDING SQ-0178 **MINOR**

Prefer `String#replaceAll()` over `String#replace()`.

Location: Backend/src/controllers/requestController.js | Line: 28

CATEGORY: STRINGS SHOULD USE "REPLACEALL()" INSTEAD OF "REPLACE()" WITH GLOBAL REGEX

SOURCE CONTEXT:

```
25 |     .replace(/&/g, "&amp;")
  26 |     .replace(/</g, "&lt;")
  27 |     .replace(/>/g, "&gt;")
  28 |     .replace(/\"/g, "&quot;")
  29 |     .replace(/'/g, "&#39;");
  30 |
  31 | const toPublicRequest = (request) => {
```

AUDIT FINDING SQ-0179 **MINOR**

Prefer `String#replaceAll()` over `String#replace()`.

Location: Backend/src/controllers/requestController.js | Line: 29

CATEGORY: STRINGS SHOULD USE "REPLACEALL()" INSTEAD OF "REPLACE()" WITH GLOBAL REGEX

SOURCE CONTEXT:

```
26 |     .replace(/</g, "&lt;")
  27 |     .replace(/>/g, "&gt;")
  28 |     .replace(/\"/g, "&quot;")
  29 |     .replace(/'/g, "&#39;");
  30 |
  31 | const toPublicRequest = (request) => {
  32 |     const publicRequest = request.toObject();
```

AUDIT FINDING SQ-0180 **MINOR**

Handle this exception or don't catch it at all.

Location: Backend/src/controllers/userController.js | Line: 792

CATEGORY: EXCEPTIONS SHOULD NOT BE IGNORED

SOURCE CONTEXT:

```
789 |             message: "Password changed successfully"
 790 |         });
 791 |
 792 |     } catch (error) {
 793 |         res.status(500).json({
 794 |             message: "Failed to change password"
 795 |         });
 796 |     }
 797 | };
 798 |
... [truncated for readability]
```

AUDIT FINDING SQ-0181 **MINOR**

Handle this exception or don't catch it at all.

Location: Backend/src/middleware/authMiddleware.js | Line: 43

CATEGORY: EXCEPTIONS SHOULD NOT BE IGNORED

SOURCE CONTEXT:

```
40 |
  41 |     next();
  42 |
  43 |     } catch (error) {
  44 |         return res.status(401).json({
  45 |             message: "Invalid or expired token"
  46 |         });
  47 |     }
  48 |
  49 | };
... [truncated for readability]
```

AUDIT FINDING SQ-0182 **MINOR**

Prefer `node:path` over `path`.

Location: Backend/src/middleware/uploadMiddleware.js | Line: 2

CATEGORY: NODE.JS BUILT-IN MODULES SHOULD BE IMPORTED USING THE "NODE:" PROTOCOL

SOURCE CONTEXT:

```
1 | const multer = require("multer");
   2 | const path = require("path");
   3 | const fs = require("fs");
   4 |
   5 | // Upload directory
```

AUDIT FINDING SQ-0183 **MINOR**

Prefer `node:fs` over `fs`.

Location: Backend/src/middleware/uploadMiddleware.js | Line: 3

CATEGORY: NODE.JS BUILT-IN MODULES SHOULD BE IMPORTED USING THE "NODE:" PROTOCOL

SOURCE CONTEXT:

```
1 | const multer = require("multer");
   2 | const path = require("path");
   3 | const fs = require("fs");
   4 |
   5 | // Upload directory
   6 | const uploadDirectory = path.join(__dirname, "../../uploads");
```

AUDIT FINDING SQ-0184 **MINOR**

`allowedMimeTypes` should be a `Set`, and use `allowedMimeTypes.has()` to check existence or non-existence.

Location: Backend/src/middleware/uploadMiddleware.js | Line: 35

CATEGORY: ARRAYS USED ONLY FOR EXISTENCE CHECKS SHOULD BE SETS

SOURCE CONTEXT:

```
32 | });
  33 |
  34 | // Allowed file types
  35 | const allowedMimeTypes = [
  36 |     "image/jpeg",
  37 |     "image/png",
  38 |     "image/webp",
```

AUDIT FINDING SQ-0185 **MINOR**

Prefer `node:fs` over `fs`.

Location: Backend/src/routes/requestRoutes.js | Line: 2

CATEGORY: NODE.JS BUILT-IN MODULES SHOULD BE IMPORTED USING THE "NODE:" PROTOCOL

SOURCE CONTEXT:

```
1 | const express = require("express");
   2 | const fs = require("fs");
   3 |
   4 |
   5 | const {
```

AUDIT FINDING SQ-0186 **MINOR**

Prefer `node:http` over `http`.

Location: Backend/src/server.js | Line: 4

CATEGORY: NODE.JS BUILT-IN MODULES SHOULD BE IMPORTED USING THE "NODE:" PROTOCOL

SOURCE CONTEXT:

```
1 | const express = require("express");
   2 | const cors = require("cors");
   3 | const helmet = require("helmet");
   4 | const http = require("http");
   5 | require("dotenv").config();
   6 |
   7 | const connectDB = require("./config/db");
```

AUDIT FINDING SQ-0187 **MINOR**

Using http protocol is insecure. Use https instead.

Location: Backend/src/services/emailService.js | Line: 4

CATEGORY: CLEAR-TEXT PROTOCOLS SHOULD NOT BE USED

SOURCE CONTEXT:

```
1 | const nodemailer = require("nodemailer");
   2 | require("dotenv").config();
   3 |
   4 | const transporter = nodemailer.createTransport({
   5 |     service: process.env.MAIL_SERVICE || "gmail",
   6 |     auth: {
   7 |         user: process.env.MAIL_USER,
```

AUDIT FINDING SQ-0188 **MINOR**

Handle this exception or don't catch it at all.

Location: Backend/src/socket/socketServer.js | Line: 58

CATEGORY: EXCEPTIONS SHOULD NOT BE IGNORED

SOURCE CONTEXT:

```
55 |                 role: user.role
  56 |             };
  57 |             return next();
  58 |         } catch (error) {
  59 |             return next(new Error("Invalid or expired token"));
  60 |         }
  61 |     });
  62 |
  63 |     io.on("connection", (socket) => {
```

AUDIT FINDING SQ-0189 **MINOR**

Remove the declaration of the unused 'formatDate' variable.

Location: Frontend/src/pages/admin/AdminRequests.jsx | Line: 263

CATEGORY: UNUSED LOCAL VARIABLES AND FUNCTIONS SHOULD BE REMOVED

SOURCE CONTEXT:

```
260 |         }
 261 |     };
 262 |
 263 |     const formatDate = (date) => {
 264 |
 265 |         if (!date) {
 266 |             return "-";
```

AUDIT FINDING SQ-0190 **MINOR**

Remove this unused import of 'FiEye'.

Location: Frontend/src/pages/auth/Register.jsx | Line: 5

CATEGORY: UNNECESSARY IMPORTS SHOULD BE REMOVED

SOURCE CONTEXT:

```
2 | import { useNavigate } from "react-router-dom";
   3 | import {
   4 |     FiBriefcase,
   5 |     FiEye,
   6 |     FiEyeOff,
   7 |     FiLock,
   8 |     FiMail,
```

AUDIT FINDING SQ-0191 **MINOR**

Remove this unused import of 'FiEyeOff'.

Location: Frontend/src/pages/auth/Register.jsx | Line: 6

CATEGORY: UNNECESSARY IMPORTS SHOULD BE REMOVED

SOURCE CONTEXT:

```
3 | import {
   4 |     FiBriefcase,
   5 |     FiEye,
   6 |     FiEyeOff,
   7 |     FiLock,
   8 |     FiMail,
   9 |     FiPhone,
```

AUDIT FINDING SQ-0192 **MINOR**

Remove this unused import of 'FiClock'.

Location: Frontend/src/pages/staff/StaffTasks.jsx | Line: 5

CATEGORY: UNNECESSARY IMPORTS SHOULD BE REMOVED

SOURCE CONTEXT:

```
2 | import {
   3 |     FiPlay,
   4 |     FiCheckCircle,
   5 |     FiClock,
   6 |     FiFileText
   7 | } from "react-icons/fi";
   8 |
```

AUDIT FINDING SQ-0193 **MINOR**

`allowedAttachmentTypes` should be a `Set`, and use `allowedAttachmentTypes.has()` to check existence or non-existence.

Location: Frontend/src/pages/user/CreateRequest.jsx | Line: 17

CATEGORY: ARRAYS USED ONLY FOR EXISTENCE CHECKS SHOULD BE SETS

SOURCE CONTEXT:

```
14 |     "General"
  15 | ];
  16 |
  17 | const allowedAttachmentTypes = [
  18 |     "image/jpeg",
  19 |     "image/png",
  20 |     "image/webp",
```

AUDIT FINDING SQ-0194 **MINOR**

Prefer `String#replaceAll()` over `String#replace()`.

Location: Frontend/src/pages/user/CreateRequest.jsx | Line: 207

CATEGORY: STRINGS SHOULD USE "REPLACEALL()" INSTEAD OF "REPLACE()" WITH GLOBAL REGEX

SOURCE CONTEXT:

```
204 |     const normalizeCategory = (value) => {
 205 |         return String(value || "")
 206 |             .toLowerCase()
 207 |             .replace(/&/g, "and")
 208 |             .replace(/[^a-z0-9\s]/g, "")
 209 |             .replace(/\s+/g, " ")
 210 |             .trim();
```

AUDIT FINDING SQ-0195 **MINOR**

Visible, non-interactive elements with click handlers must have at least one keyboard listener.

Location: Frontend/src/pages/user/MyRequests.jsx | Line: 364

CATEGORY: MOUSE EVENTS SHOULD HAVE CORRESPONDING KEYBOARD EVENTS

SOURCE CONTEXT:

```
361 |
 362 |                                                 <div className="request-info">
 363 |
 364 |                                                     <strong
 365 |                                                            className="request-
clickable"
 366 |                                                            onClick={() =>
 367 |                                                                navigate(`/user/
requests/${request._id}`)
 368 |                                                            }
 369 |                                                       >
 370 |                                                            {request.title}
... [truncated for readability]
```

3. Remediation Roadmap

Based on the automated assessment, the following actions are recommended to bring the codebase up to Enterprise Standards:

%ï Short-term:
Eliminate all BLOCKER and CRITICAL severity issues. These represent immediate risks to system stability.

%ï Security:
Review all vulnerabilities and ensure sensitive data is not logged, exposed, or transmitted insecurely.

%ï Quality:
Increase unit test coverage to at least 80%. Automated tests prevent regression and improve reliability.

%ï Technical:
Refactor high-complexity functions. Smaller, focused modules are easier to maintain and audit.

Disclaimer: This report is generated based on automated static analysis and should be reviewed by a lead developer.